

ANOTHER LINEAR DIAGONALS-PARAMETER SYMMETRY MODEL FOR SQUARE CONTINGENCY TABLES WITH ORDERED CATEGORIES

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Summary: For square contingency tables with ordered categories, Agresti (1983) considered the linear diagonals-parameter symmetry (LDPS) model. This paper proposes another LDPS model in which the positive (negative) log odds parameters have a decreasing (increasing) linear pattern of the distance from the diagonal. The proposed model is a special case of Tomizawa's (1987) 2-ratios-parameter symmetry model, and also a simple version of Tomizawa's (1984, 1987) extended quasi-symmetry model. The model is applied to the British father-son occupational mobility data.

1. INTRODUCTION

For the $R \times R$ square contingency table with ordered categories, let p_{ij} denote the probability that an observation will fall in the cell in row i and column j ($i = 1, 2, \dots, R; j = 1, 2, \dots, R$). Goodman (1979) considered the diagonals-parameter symmetry (DPS) model, defined by

$$p_{ij} = \begin{cases} \delta_{j-i}\psi_{ij} & \text{for } i < j, \\ \psi_{ij} & \text{for } i \geq j, \end{cases}$$

where $\psi_{ij} = \psi_{ji}$. Special cases of this model are given in Table 1 (for details, see Agresti, 1984, Chap. 11; Tomizawa, 1987).

Table 1
Special cases of the DPS model applied to the $R \times R$ table

Models	Restrictions imposed on parameters $\{\delta_k\}, k = 1, 2, \dots, R-1$
Symmetry (S)	$\delta_k = 1$
Conditional symmetry (CS)	$\delta_k = \tau$
Linear diagonals-parameter symmetry (LDPS)	$\delta_k = \rho^k$
2-ratios-parameter symmetry (2RPS)	$\delta_k = \phi\theta^{k-1}$
Another LDPS (ALDPS)	$\delta_k = \delta^{R-k}$

Under the LDPS model (with $\rho \neq 1$), the odds $\{p_{ij}/p_{ji}\}$ increases (decreases) monotonically for $i < j$ with distance $j - i$ ($j - i = 1, 2, \dots, R - 1$) from the diagonal of the $R \times R$ table whenever an arbitrary odds is greater (less) than one. Under the 2RPS model (with $\phi \neq \theta$ and $\theta \neq 1$), the odds always increases or decreases monotonically with the distance from the diagonal without depending on whether an arbitrary odds is greater or less than one. We are now interested in a model under which the odds increases (decreases) monotonically with the distance from the diagonal whenever an arbitrary odds is less (greater) than one.

The purpose of this paper is to propose such a model which is applied to the British father-son occupational mobility data.

2. ANOTHER LDPS MODEL

Consider a model defined by

$$p_{ij} = \begin{cases} \delta^{R-(j-i)} \psi_{ij} & \text{for } i < j, \\ \psi_{ij} & \text{for } i \geq j, \end{cases}$$

where $\psi_{ij} = \psi_{ji}$. Under this model, the log odds that an observation will fall in one of the cells with a certain distance above the diagonal of the $R \times R$ table, rather than in one of the cells with the same distance below it, is a linear function of the distance with slope $(-\log \delta)$. Therefore we will refer to this model as another LDPS (ALDPS) model. Under the ALDPS model, the above odds increases (decreases) monotonically with the distance from the diagonal whenever it is less (greater) than one.

Let X and Y denote the row and column variables, respectively. Under the ALDPS model, $\delta \geq 1$ is equivalent to $Pr(X \leq i) \geq Pr(Y \leq i)$ for every i . Therefore the ALDPS model would be useful for making inferences such as that X is stochastically less than Y or vice versa.

Caussinus (1965) considered the quasi-symmetry (QS) model and Tomizawa (1984, 1987) considered an extended QS (EQS) model, defined by

$$p_{ij} = \begin{cases} \Delta \alpha_i \beta_j \psi_{ij} & \text{for } i < j, \\ \alpha_i \beta_j \psi_{ij} & \text{for } i \geq j, \end{cases}$$

where $\psi_{ij} = \psi_{ji}$. The EQS model is also equivalent to

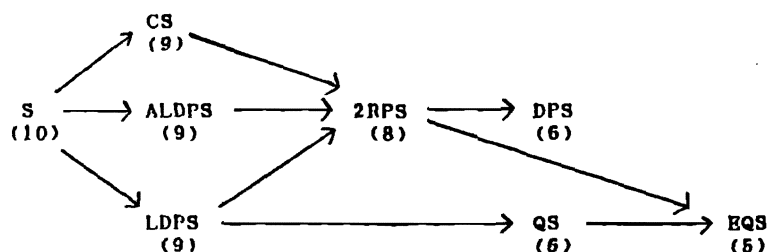
$$(2.1) \quad \frac{p_{ij}}{p_{ji}} = \Delta \frac{\nu_j}{\nu_i} \quad \text{for } 1 \leq i < j \leq R,$$

where $\nu_1 = 1$ without loss of generality. The EQS model with $\Delta = 1$ is the QS model. The models described in Table 1 are special cases of the EQS model (see Table 2 and Figure 1).

Table 2
Special cases of the EQS model with form (2.1)

Models	Restrictions imposed on parameters
S	$\Delta = 1, \quad \nu_1 = \nu_2 = \dots = \nu_R (= 1)$
CS	$\nu_1 = \nu_2 = \dots = \nu_R (= 1)$
LDPS	$\Delta = 1, \quad \nu_i = \nu^{i-1} (i = 1, 2, \dots, R)$
ALDPS	$\nu_i = \Delta^{-(i-1)/R} (i = 1, 2, \dots, R)$
2RPS	$\nu_i = \nu^{i-1} (i = 1, 2, \dots, R)$
QS	$\Delta = 1$

Figure 1
Hierarchical tree for models.



Note: $A \rightarrow B$ indicates that model A implies model B . The number in parenthesis beside each model is the associated number of df for the data in Table 3.

The LDPS model is a simple version of the QS model, however, the ALDPS model is not so, instead a simple version of the EQS model. (Note that in Figure 1 the S and QS models treat classifications as nominal and the other models treat them as ordinal.)

Table 3

Occupational status for British father-son pairs; from Agresti (1984, p. 206). (The upper and lower parenthesized values are the maximum likelihood estimates of expected frequencies under the CS model and ALDPS model, respectively.)

Father's status	Son's status					Total
	(1)	(2)	(3)	(4)	(5)	
(1)	50 (50) (50)	45 (40.7) (41.1)	8 (10.6) (10.4)	18 (17.8) (17.0)	8 (6.1) (5.7)	129
(2)	28 (32.3) (31.9)	174 (174) (174)	84 (90.3) (91.2)	154 (169.5) (166.4)	55 (54.1) (51.6)	495
(3)	11 (8.4) (8.6)	78 (71.7) (70.8)	110 (110) (110)	223 (227.5) (229.7)	96 (93.7) (91.9)	518
(4)	14 (14.2) (15.0)	150 (134.5) (137.6)	185 (180.5) (178.3)	714 (714) (714)	447 (427.7) (431.8)	1510
(5)	3 (4.9) (5.3)	42 (42.9) (45.4)	72 (74.3) (76.1)	320 (339.3) (335.2)	411 (411) (411)	848
Total	106	489	459	1429	1017	3500

The ALDPS model can be tested for goodness of fit by the likelihood ratio statistic G^2 with $(R + 1)(R - 2)/2$ degrees of freedom (df) which equals df for the CS and LDPS models.

3. OCCUPATIONAL MOBILITY EXAMPLE

Table 3 taken from Agresti (1984, p.206) is the father's and son's occupational mobility data in Britain. From Table 4, the LDPS model fits these data poorly, however, the 2RPS model fits the data well. From Table 5, the pattern of increase or decrease for the estimated odds under the LDPS model is opposite to that under the 2RPS model. This illustrates why the LDPS model fits these data poorly. Also, the pattern of the estimated odds under the 2RPS model suggests that the ALDPS model rather than the LDPS model would tend to fit these data better. In fact, the ALDPS model fits these data well.

Table 4

Likelihood ratio statistics G^2 and the AIC^+ values for models applied to the data in Table 3

Applied models	DF	G^2	AIC^+
S	10	37.46**	17.46
CS	9	10.35	- 7.65
LDPS	9	17.13*	- 0.87
ALDPS	9	10.13	- 7.87
2RPS	8	10.02	- 5.98
DPS	6	6.44	- 5.56
QS	6	4.66	- 7.34
EQS	5	2.70	- 7.30

Note: $AIC^+ = G^2 - 2$ (number of df).

*(**) means significant at 5 % (1 %) level.

From Table 5, the ALDPS model provides information that the proportion of father-son pairs for which the son had the k grades higher status category for $k = 1, 2, 3, 4$, is estimated to be $(1.065)^{5-k}$ times higher than the proportion in which the father had the k grades higher status category. In addition, since the estimated odds are greater than one under the ALDPS model, the son's occupational status tends to be stochastically higher than the father's.

According to the tests based on the difference between the G^2 values for nested two models (though they are omitted), the ALDPS and CS models are preferable to the 2RPS, DPS and EQS models for these data. The comparison of the ALDPS and CS models will be done in the next section.

Table 5

Estimated values of odds $\{p_{ij}/p_{ji}\}$ for $j - i = 1, 2, 3, 4$ under the LDPS, ALDPS and 2RPS models applied to the data in Table 3

Models	$j - i = 1$	2	3	4
LDPS	1.141	1.302	1.486	1.696
ALDPS	1.288	1.209	1.135	1.065
2RPS	1.280	1.229	1.181	1.134

4. MODEL SELECTION BY AKAIKE'S INFORMATION CRITERION

We shall select the best-fitting model by using the Akaike's (1974) information criterion (AIC) defined as

$$AIC = -2(\text{maximum log-likelihood}) + 2(\text{number of free parameters})$$

for each model. This criterion gives the best-fitting model as the one with minimum AIC. Since we need only the difference between AIC's when two models are compared, we ignore the common constant and use a modified AIC defined as

$$AIC^+ = G^2 - 2(\text{number of degrees of freedom}).$$

From Table 4, the ALDPS model is the best-fitting model for the data in Table 3 by this criterion. Also the second best-fitting model is the CS model. However, the AIC^+ values for both models seem close. Therefore, by the AIC criterion, the ALDPS model is better than or almost similar to the CS model for these data.

5. ANALYSIS OF RESIDUALS

Table 6 gives the standardized residuals for the CS and ALDPS models applied to the data in Table 3. For both models, the deviation for cell (4,2)

is somewhat greater than those for other cells, and also it is greater for the CS model than for the ALDPS model. The pattern of positive or negative deviations is the same for both models. Especially the last row and column [except for cell (5,5)] give all negative and positive residuals, respectively. Therefore the observations show that the proportion of father-son pairs for which the son (father) had the highest status category "5" tends to be greater (less) than would be expected under each of the CS and ALDPS models.

Table 6

Standardized residuals for the CS (upper entry) and ALDPS (lower entry) models applied to the data in Table 3

Father's status	Son's status				
	(1)	(2)	(3)	(4)	(5)
(1)	0	0.67	-0.80	0.05	0.77
	0	0.61	-0.74	0.24	0.96
(2)	-0.76	0	-0.66	-1.19	0.12
	-0.69	0	-0.75	-0.96	0.47
(3)	0.90	0.74	0	-0.30	0.24
	0.82	0.86	0	-0.44	0.43
(4)	-0.05	1.34	0.33	0	0.93
	-0.26	1.06	0.50	0	0.73
(5)	-0.86	-0.14	-0.27	-1.05	0
	-1.00	-0.50	-0.47	-0.83	0

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